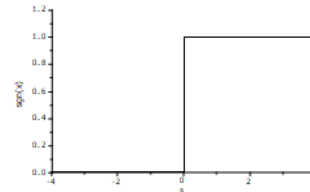


A function $y = f(x)$ describes a relationship (input-output mapping) from x to y .

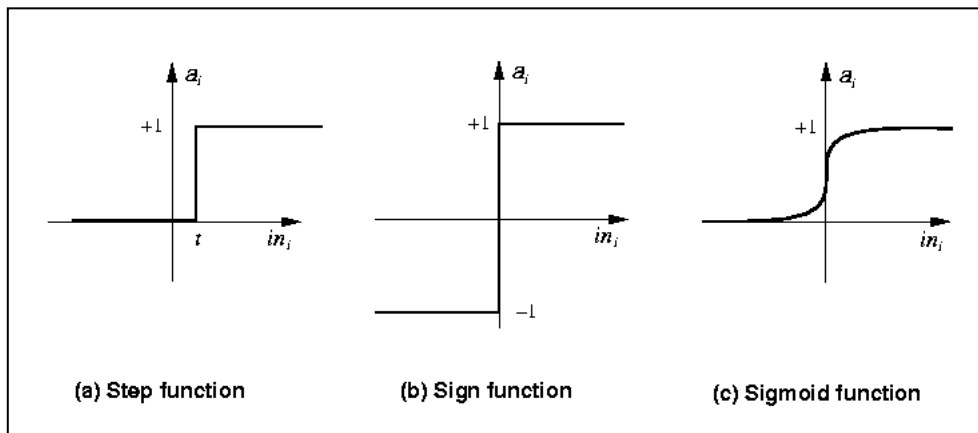
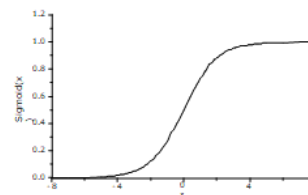
Example 1 The threshold or step function $\text{step}(x)$ is defined as

$$\text{step}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$



Example 2 The logistic sigmoid function $\text{Sigmoid}(x)$ is defined as

$$\text{Sigmoid}(x) = \frac{1}{1 + e^{-\lambda x}}$$



activation function. Typical activation functions used are

$$f(net) \triangleq \frac{2}{1 + \exp(-\lambda net)} - 1 \quad (2.3a)$$

and

$$f(net) \triangleq \text{sgn}(net) = \begin{cases} +1, & net > 0 \\ -1, & net < 0 \end{cases} \quad (2.3b)$$

where $\lambda > 0$ in (2.3a) is proportional to the neuron gain determining the steepness of the continuous function $f(net)$ near $net = 0$. The continuous activation function is shown in Figure 2.5(a) for various λ . Notice that as $\lambda \rightarrow \infty$, the limit of the continuous function becomes the $\text{sgn}(net)$ function defined in (2.3b). Activation functions (2.3a) and (2.3b) are called *bipolar continuous* and *bipolar binary functions*, respectively. The word “bipolar” is used to point out that both positive and negative responses of neurons are produced for this definition of the activation function.

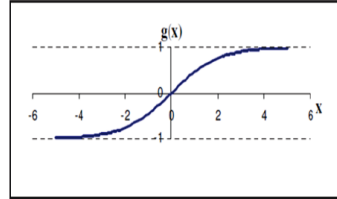


FIGURE 2. Graphical representation of Bipolar sigmoid activation function

In order to compute the output, we use the value of local field by applying equation (2) into equation (5).

$$y(x) = g(h_i) \quad (5)$$

Bipolar sigmoid activation function is given by

$$g(h_i) = \frac{1 - e^{-h_i}}{1 + e^{-h_i}} \quad (6)$$